

Project

Linear Additive models – Calibration, Monte Carlo, Exotics and their RM Financial Engineering

You are asked to consider three different models: the Minimal Additive (MA) model and two benchmark models: the Generalized Logistic (GL) with real valued underlying and the Additive Bachelier (AB). Calibrate from the European options on WTI oil options prices on the 2nd of June 2020; simulate the processes and price/RM exotics.

The folder “Data” contains

- The folders “datacalls” and “dataputs”, that contain the files with the mid prices of call (and put) options for the different maturity (the filename corresponds to the option’s expiry, each file contains prices on all the grid of strikes for each value date).
- The file “Expiries_Futures”, with the expiry date of each option.

Consider the 3 models: the minimal Additive model; the GL model with real valued underlying, the AB model (characteristic function in (2) of [3]), with $\alpha=1/2$.

0) Asymptotic properties

exponential tails for a given time horizon

- a. Deduce the asymptotic exponential tail coefficients of $f_t(p_t^+, p_t^-)$
- b. Plot in a log-linear plot the pdf of the three models with same $p_t^+ = 1.5, p_t^- = 0.9$

1) Calibration Forwards and Discount Factors

Calibrate Forwards and Discount Factors following the technique described in [2]. Hereinafter, based on these calibrated values, consider zero-rates and Absolute dividends as deterministic functions of time.

2) Calibrate the Implied Volatility Surface (Consider Fwd-OTM Calls and Puts)

- a. Pricing formula for Linear Additive processes for European Call in [3].
- b. AB as in [3] using with initial guess for the parameters $\eta_0 = 0.2$ and $k_0 = 1$.
- c. Consider MA and GL models if α_t, β_t do not depend on time. Is it possible the cascade calibration described in [3, Sec.3] also for these models?
- d. Compare the results.

3) Simulation (between intermediate reset dates $t > s$)

- a. Compute the CF between any pair of reset dates $t > s \geq t_0$, using the additivity property for the driving process $\{x_t\}_t$ that the three models satisfy
$$E_0[e^{i x_t u}] = E_0[e^{i (x_t - x_s) u}] E_0[e^{i x_s u}].$$
- b. Compute the (conditional) $CDF(x_t | x_s)$ for the increments via Fourier Transform (similarly to the one when pricing a Digital option).
- c. Simulate the paths (between reset dates in $T_1 = 6m, T_2 = 1y$) using the Lewis-FFT algorithm (the steps are described in Appendix B of [1] and [5]).

Hints:

i) In the FFT consider only the part of the CF coming from the absolutely continuous part of the pdf (cf. also [5]). Why?

ii) Pay attention to the relations between forward (Forwards doc).

- d. Verify the adequacy of the MC on the forward start option with $T_1 = 6m, T_2 = 1y$ for some K_2 , whose payoff in T_2 is

$$[S_{T_2} - K_2 F(T_1, T_2)]^+$$

comparing the MC price with the analytical price when available.

- e. [Facultative] In the MA model, is it possible to compute the $CDF(x_t | x_s)$ analytically? In that case, compare computational time and precision of the MC e.g. considering the first 4 moments.

4) Pricing exotic payoffs.

Consider the following

- a. compound option: a call-on-call (CoC) and a put-on-put (PoP); the CoC reads

$$CoC(t_0; T_1, K_1; T_2, K_2) = B(t_0, T_1)E_0[B(T_1, T_2)E_1[S(T_2) - K_2]^+ - K_1]^+$$

- b. chooser option

$$Chooser(t_0; T_1; T_2, K_2) = B(t_0, T_2)E_0[\max(E_1[S(T_2) - K_2]^+, E_1[K_2 - S(T_2)]^+)]$$

The strike K_2 is equal to $F(T_0, T_2)$. Price the exotic option with the three models for some K_1 .

Hint: MC pricing with the 3 models. Simulate the paths using the Lewis-FFT algorithm. The market maker quotes these exotics with max 10 bps of bid-ask. Check if a “smart extrapolation” of the CDF outside the FFT range (cf also point 0) and [5]) can improve the simulation when the number of simulation points in the FFT grid is “not-so-large”.

- 5) [Facultative] For the MA, is it possible to obtain an elementary closed formula for the exotics in 4)? Compare the prices.

- 6) [Facultative] Risk Management.

Hedge the products with plain vanilla options and the underlying front future (establish a quantitative rule). The hedging cost (bid-ask) is 1bp for the underlying and 4bp for each PV option. Check the quality of the hedge on the next two Tuesdays: control the total hedging cost and the P&L.

References

- [1] Azzone, M., & Baviera, R. (2023). A fast Monte Carlo scheme for additive processes and option pricing. *Computational Management Science*, 20(1), 31. Available at <https://doi.org/10.1007/s10287-023-00463-1>
- [2] Azzone, M., & Baviera, R. (2021). Synthetic forwards and cost of funding in the equity derivative market. *Finance Research Letters*, 41, 101841.
- [3] Baviera, R., & Massaria, M.D. (2026). The Additive Bachelier model with an application to the oil option market in the Covid period, *Journal of Computational and Applied Mathematics*, 487, 117741. Available at <https://doi.org/10.1016/j.cam.2026.117741>
- [4] Carr, P., & Torricelli, L. (2021). Additive logistic processes in option pricing. *Finance and Stochastics*, 25(4), 689-724.
- [5] Baviera, R., & Manzoni, P. (2026). Fast and General Simulation of Lévy-driven OU processes for Energy Derivatives. *arXiv preprint arXiv:2401.15483*.

Realize a library in Matlab.

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